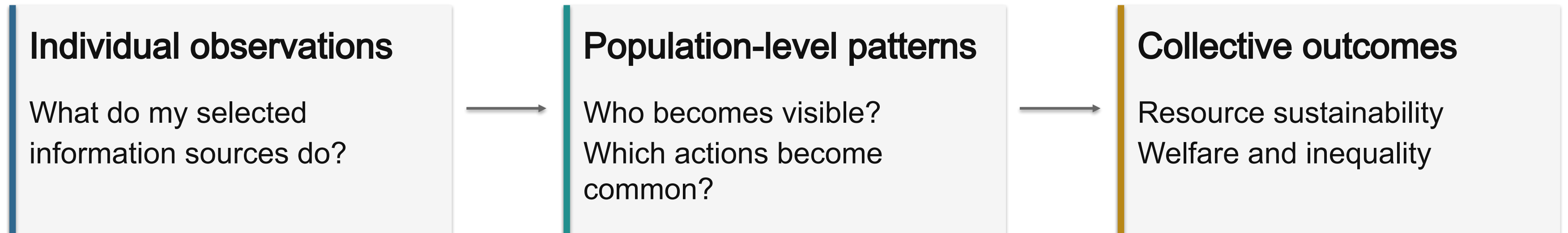


How do local observations become collective outcomes?

Can changing whom agents observe produce population-level patterns that sustain a shared renewable resource?

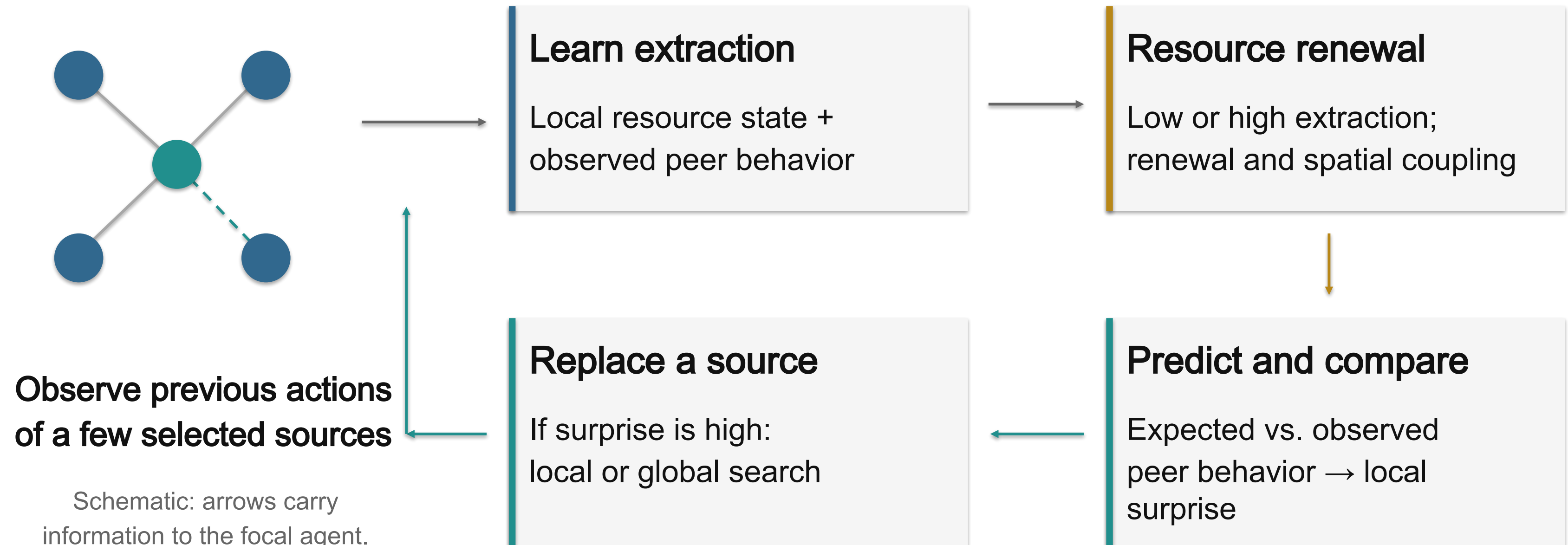


HYPOTHESIS

Prediction-error-driven rewiring produces patterns that improve sustainability beyond the same amount of random network turnover.

One feedback loop connects agents, networks, and ecology

Agents stay in place. Information links can change; resources do not travel along them.



Reward = own realized harvest. Sustainability is measured, not rewarded.

The key test holds the amount of rewiring constant

Adaptive rewiring versus matched random turnover, within the same ecological condition.

Three reference conditions

- B0** Ecological learning only
- S1** Fixed random observation network
- S2** Fixed network with uneven visibility

R1 / R2 / R3 ↔ matched R0

Local / mixed / global search

Same event counts at every checkpoint

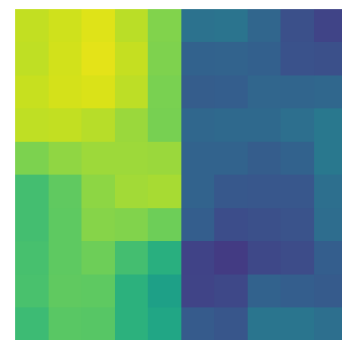
Different rule for which observers rewire



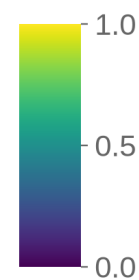
Uniform high



Patchy high



Split high / low



3 ecologies × 2 populations (32, 64)

10 replicates per condition

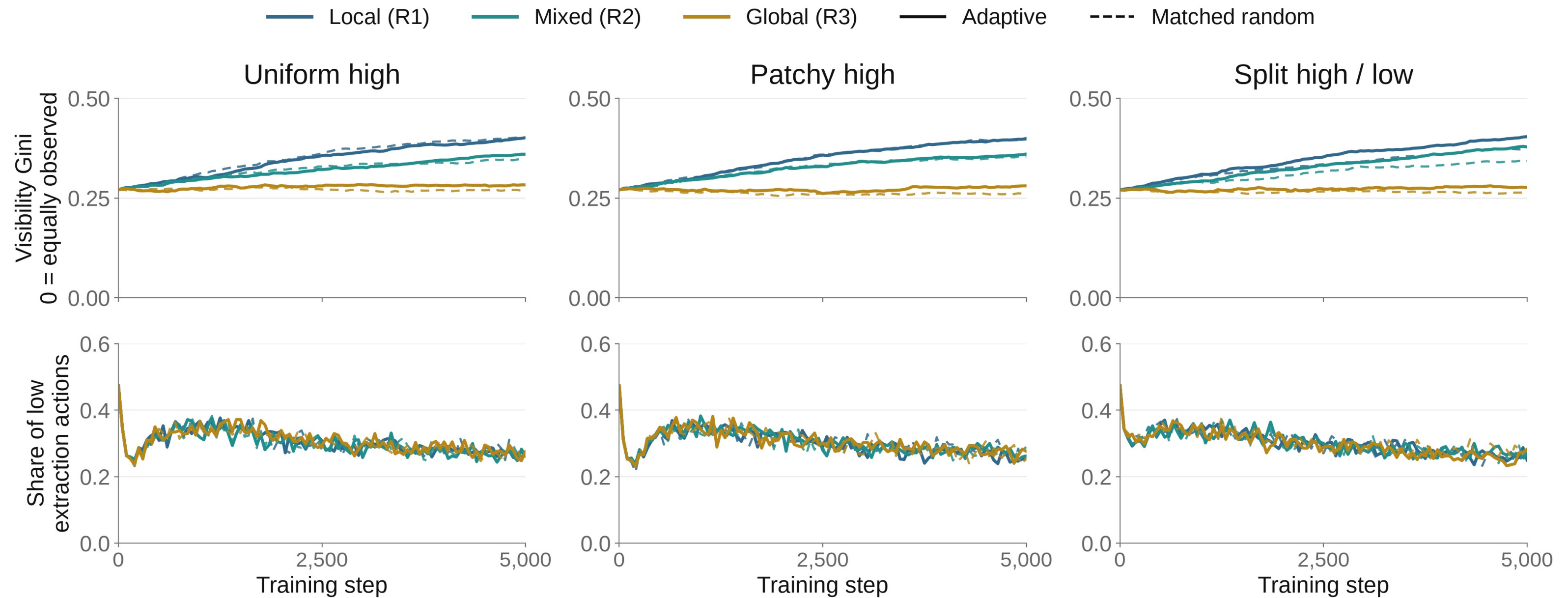
5,000 training + 1,000 evaluation steps

Fresh ecology; learned policies and graphs frozen

Capacity maps · replicate 0 · common scale 0–1

Local search concentrates attention—even with random turnover

Population-level patterns during training · N = 64 · all three ecologies

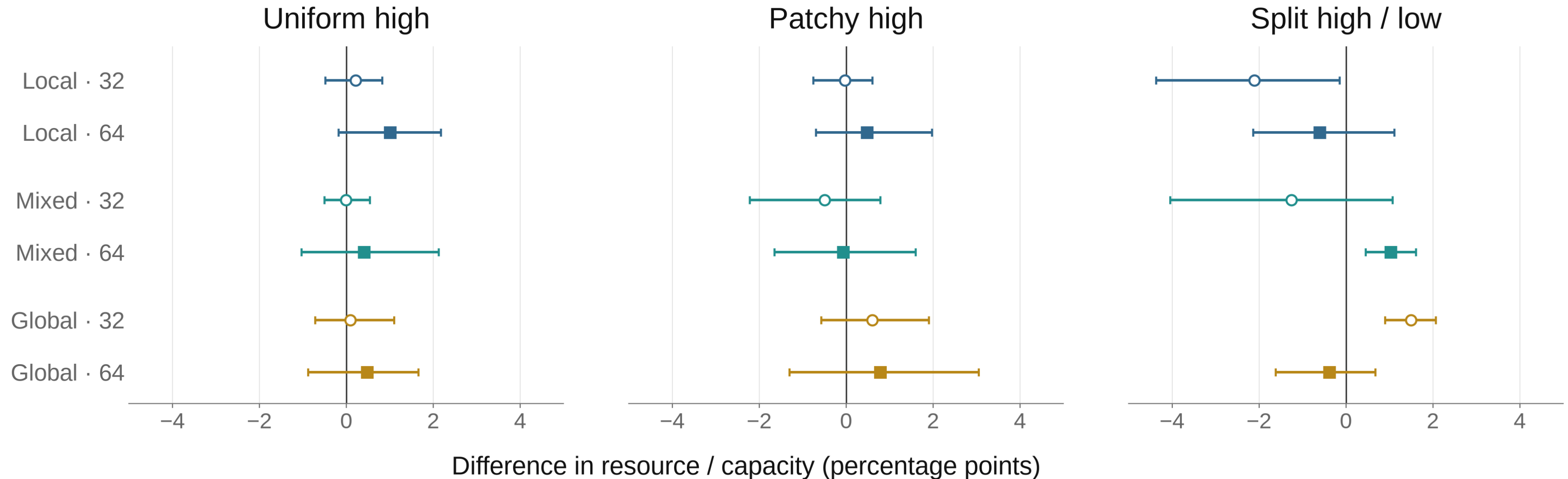


Organization appears in both adaptive and matched-random networks.

Lines: means of 10 replicates, unsmoothed; descriptive trends, not uncertainty intervals. N=32 companion supplied.

Adaptive rewiring has no clear general sustainability advantage

Difference from matched random turnover · every ecology, search scope, and population

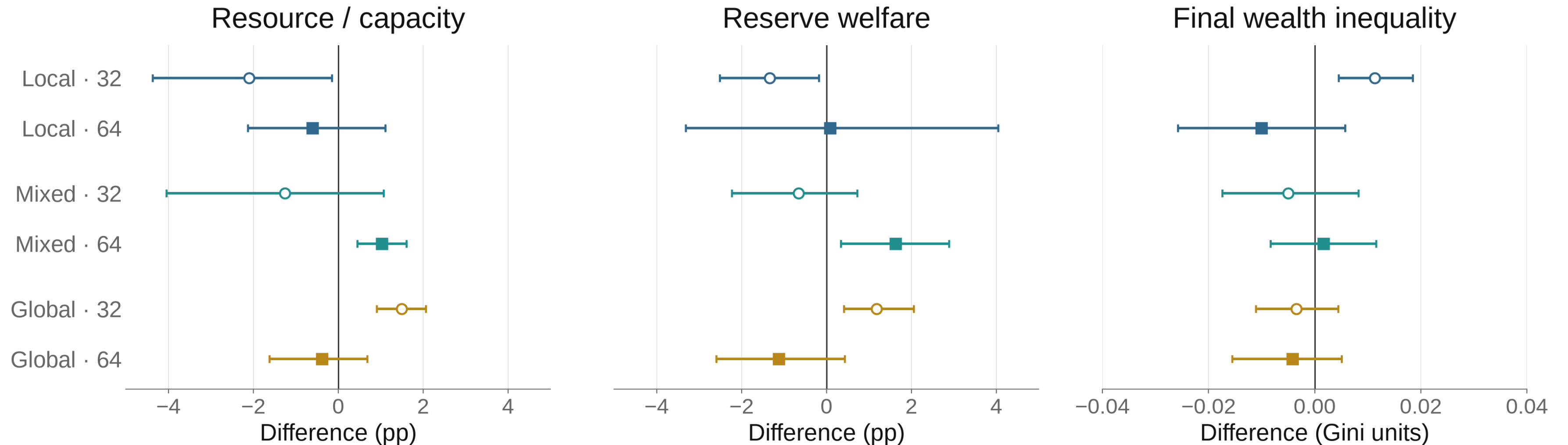


15 of 18 intervals cross zero; the direction also changes across conditions.

Right of zero = more resource under adaptive rewiring. Rows show search scope and population.
10 paired replicates; 95% bootstrap intervals (2,000 resamples), exploratory and not multiplicity-adjusted.

A resource gain is not a complete welfare story

Closer look: split high / low ecology · all six adaptive-versus-R0 comparisons



Mixed search, N=64: resource +1.02 pp; reserve welfare +1.62 pp.

Its wealth-inequality interval crosses zero. More resource does not establish a fairer outcome.

Positive = more resource / welfare, but more inequality. Same 10 pairs and 95% bootstrap method as slide 5.

The project can stay focused on one micro-to-macro test

Individual observation → population patterns → sustainability, welfare, and inequality

What the pilot shows

Local search changes
who becomes visible.

Random turnover can produce
a similar concentration.

What remains unresolved

Does the error trigger create
a distinct collective pattern?

Does that pattern improve
resource outcomes?

What the next test needs

Specify the expected
population pattern.

Keep matched R0 and
replicate-level uncertainty.

Feedback I want from you

Is the question narrow enough?

Is “collective organization” defined clearly enough to test?

Which link in the mechanism needs the strongest evidence?